(a) Conveyor sorting

illustration: side view
overhead camera + FOV cone
belt with boxes, target box tracked
ghost box at t+Δ (predicted)
arm intercepts on the move

1. Track box, estimate $\hat{v}$
2. Predict pose at $t+\Delta$, plan grasp
3. Sync with belt, grasp on the move

(b) Catching a moving object

illustration: side view
observed ball samples (fading dots)
predicted parabola (dashed)
catch point p^* + uncertainty ellipse
ghost start pose → catch pose

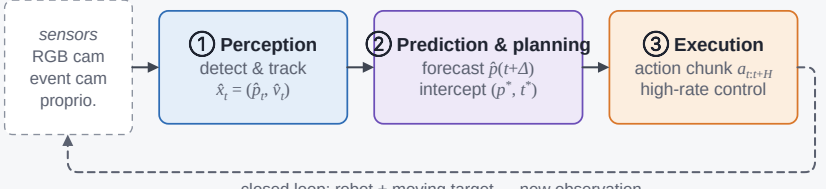
1. Detect ball, fit trajectory
2. Predict catch point (p^*, t^*)
3. Reach p^* before t^*

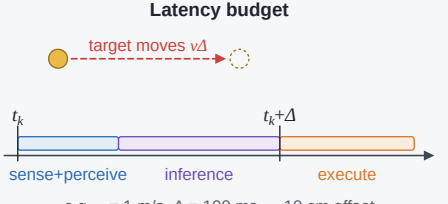
(c) Table-tennis striking

illustration: side view
event camera, table + net
incoming ball, bounce, hit point
racket arm: back-swing → strike
swing arc (orange)

1. High-rate ball tracking (events)
2. Predict bounce & hitting point
3. Timed swing, velocity profile

(d) Closed loop under latency: perception → prediction & planning → execution





e.g. v = 1 m/s, Δ = 100 ms → 10 cm offset
⇒ act on the predicted state, not the last observation